# Complete Mathematical Formulation: M006 Clamped (Optimized MAGI)

The **M006 Clamped** architecture (also referred to as the Kernelized Symplectic-HDDM or Optimized MAGI) represents a biologically plausible cognitive model bridging deep cerebellar reservoir computing with the cortical Hierarchical Drift-Diffusion Model (HDDM).

This document strictly defines the exact mathematical sequence evaluated for every subject trial $t$. All boundaries are governed by continuously differentiable algebraic squashing (using tanh and $\text{logit}^{-1}$) to guarantee a strict, unbroken manifold for Hamiltonian Monte Carlo (HMC) sampling without Jacobian singularities.

## 1. Cortical Action Value Tracking (The Cortex)

The cortex maintains an action value $Q_c^{(t)}$ for choice $c \in \{1, 2\}$, updated via a standard Rescorla-Wagner learning rule on trial $t$. * **Prediction Error**: $E_t = R_t - Q_{c_t}^{(t)}$ * **Action Value Update**: $Q_{c_t}^{(t+1)} = Q_{c_t}^{(t)} + \alpha_{\text{ctx}} E_t$ * **Cortical Action Gradient**: The difference in action values is tracked as an integrated scalar:

$$Q_{\text{diff}}^{(t)} = Q_2^{(t)} - Q_1^{(t)}$$

## 2. Cerebellar Temporal Extension (Mossy Fiber Integration)

The cerebellum observes the cortical action values $Q_{c_t}^{(t)}$ through a set of random, fixed topological projections $\mathbf{W}_{\text{exp}} \in \mathbb{R}^{32}$. It tracks them via leaky integrators $\mathbf{M}^{(t)} \in \mathbb{R}^{32}$, where each temporal node maintains a distinct decay scale. * **Memory Trace Update**:

$$\mathbf{M}^{(t)} = \boldsymbol{\alpha}_{\text{frac}} \odot \mathbf{M}^{(t-1)} + (\mathbf{1} - \boldsymbol{\alpha}_{\text{frac}}) \odot \left( \mathbf{W}_{\text{exp}} \times Q_{c_t}^{(t)} \right)$$

*Where* $\boldsymbol{\alpha}_{\text{frac}} \in \mathbb{R}^{32}$ is a uniform gradient spanning $[0.1, 0.9]$.

## 3. Purkinje Cell Reservoir Dynamics

The 32-node Purkinje population state $\mathbf{Z}^{(t)} \in \mathbb{R}^{32}$ dynamically integrates the mossy fiber traces with a biophysical, ITI-dependent (Inter-Trial Interval) exponential decay. * **Biological Decay Factor**: $\phi_{\text{decay}}^{(t)} = \exp\left(-\frac{\text{ITI}_t}{\tau_{\text{decay}}}\right)$ * **Purkinje State**:

$$\mathbf{Z}^{(t)} = \phi_{\text{decay}}^{(t)} (\boldsymbol{\kappa} \odot \mathbf{Z}^{(t-1)}) + \tanh(\mathbf{M}^{(t)})$$

*Where* $\boldsymbol{\kappa} \in \mathbb{R}^{32}$ is a retention constant scaling linearly from 0.1 to 0.99.

## 4. Synaptic Weight Matrix & Error Tracking

The Purkinje-to-Deep Cerebellar Nuclei (DCN) weights $\mathbf{W}_{\text{PC}} \in \mathbb{R}^{32}$ associate specific spatial Purkinje patterns with incoming cortical prediction errors. * **Latent Weight Update**:

$$\mathbf{W}_{\text{PC, latent}}^{(t+1)} = \mathbf{W}_{\text{PC, latent}}^{(t)} + \alpha_{\text{pc}} E_t \mathbf{Z}^{(t)}$$

* **Effective Clamped Weights**: To prevent asymptotic divergence in HMC, the latent weights are strictly bounded using a scaled hyperbolic tangent mapping:

$$\mathbf{W}_{\text{eff}}^{(t)} = 3.0 \times \tanh\left( \frac{\mathbf{W}_{\text{PC, latent}}^{(t)}}{3.0} \right)$$

## 5. Symplectic Rebound & Golgi Sparsification

The effective Purkinje signal evaluates the alignment between current reservoir patterns and learned weights. A Golgi-cell-like inhibitory mask assesses spatial entropy. * **Effective Trace**: $\mathbf{E}_{\mathbf{Z}}^{(t)} = \mathbf{W}_{\text{eff}}^{(t)} \odot \mathbf{Z}^{(t)}$ * **Differentiable Sparsity Mask**:

$$\mathbf{S}^{(t)} = \tanh\left(\gamma_{\text{golgi}}\sqrt{\mathbf{E}_{\mathbf{Z}}^{(t)} \odot \mathbf{E}_{\mathbf{Z}}^{(t)} + 10^{-8}}\right)$$

* **Bilateral Rebound Consolidation**: The 32 nodes are pooled into two opposing streams (representing competing cerebellar microzones):

$$CB_0^{(t)} = \mathbf{S}_{1:16}^{(t)} \cdot \mathbf{E}_{\mathbf{Z},1:16}^{(t)}$$

$$CB_1^{(t)} = \mathbf{S}_{17:32}^{(t)} \cdot \mathbf{E}_{\mathbf{Z},17:32}^{(t)}$$

## 6. Neuromodulatory DDM Formulation (Dynamic Gating)

The cerebellum outputs its spatial entropy metric to the Locus Coeruleus (LC) and Basal Ganglia to dynamically gate the Drift-Diffusion bounds.

### 6.1. Dynamic Drift Rate ($v_t$)

Drift rate represents momentary directional evidence, driven by the cortical value gradient and cerebellar temporal foresight. It is bounded to ±18.51 (empirical physiological bound). * **Scaled Drift Velocity**:

$$v_{\text{eff}}^{(t)} = v_{\text{ctx}} Q_{\text{diff}}^{(t)} + \gamma_{\text{var}}\left(CB_1^{(t)} - CB_0^{(t)}\right)$$

* **Clamped Drift Rate**:

$$v_t = 18.51 \times \tanh(v_{\text{eff}}^{(t)} \times 0.0540248)$$

### 6.2. Dynamic Boundary Separation ($a_t$)

Boundary separation tracks Epistemic Deliberation. When both microzones exhibit high structural entropy (large magnitudes of opposing errors tracked across different timescales), the model detects complexity and triggers a deliberative state by expanding the decision boundary. If entropy is low, the boundary collapses, enabling a fast Win-Stay/Lose-Shift heuristic. * **Raw Boundary Space**:

$$a_{\text{raw}}^{(t)} = a_{\text{base}} + w_u\sqrt{\left((CB_0^{(t)})^2 + 10^{-8}\right)\left((CB_1^{(t)})^2 + 10^{-8}\right)}$$

* **Clamped Boundary Mapping**:

$$a_t = 0.11 + 7.36 \times \text{logit}^{-1}(a_{\text{raw}}^{(t)})$$

## 7. Joint Likelihood (Wiener Diffusion)

The final trial behavioral prediction (Reaction Time and Choice) is evaluated probabilistically through the Wiener First-Passage Time distribution (assuming starting bias $w = 0.5$).

$$RT_t \sim \text{Wiener}(a_t,\, tnd,\, 0.5,\, v_t)$$